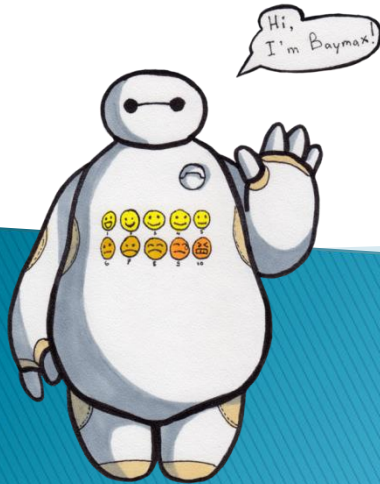


COMP1008 - Fundamentals of AI

AI in the School and in the Wild

– What's next?

Professor Rong Qu
rong.qu@nottingham.ac.uk



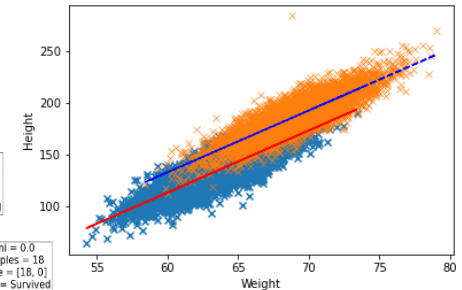
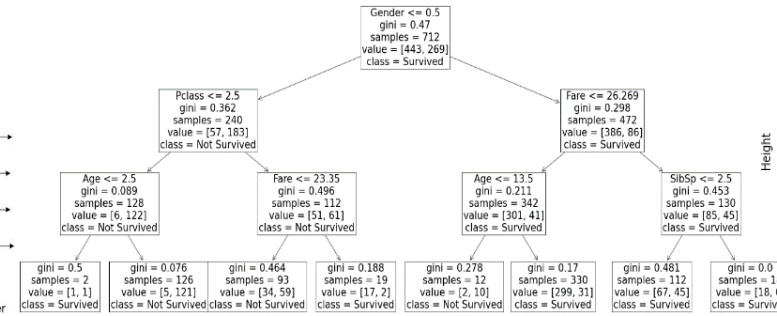
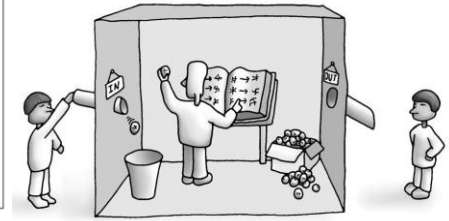
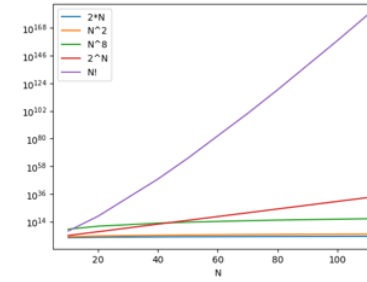
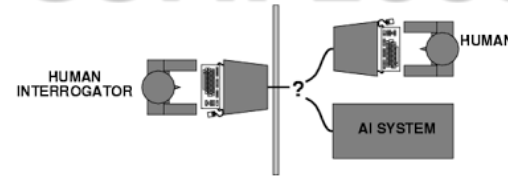
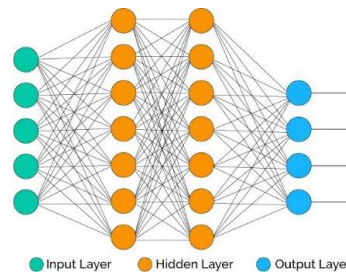
AI @ the School – COMP1008

Part I: Fundamental Issues

- Combinatorial explosion, experiments

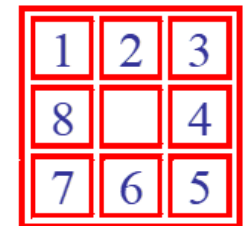
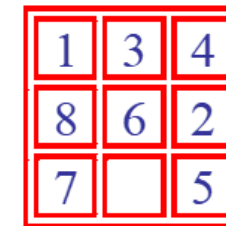
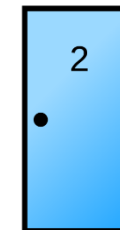
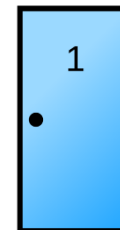
Part II: Machine learning

- Artificial Neural Networks
- Regression
- Decision Tree



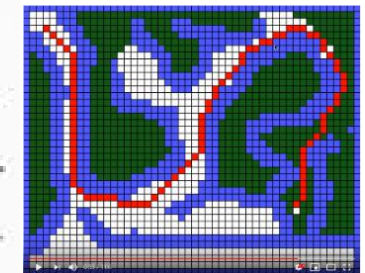
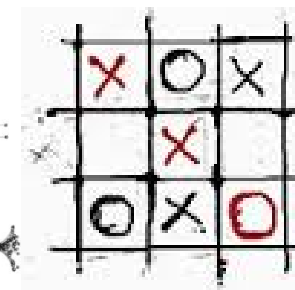
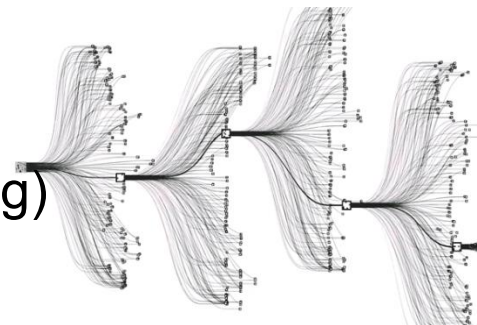
Part III: Intelligent Systems

- KRR, Bayes Rule
- NLP, LLMs



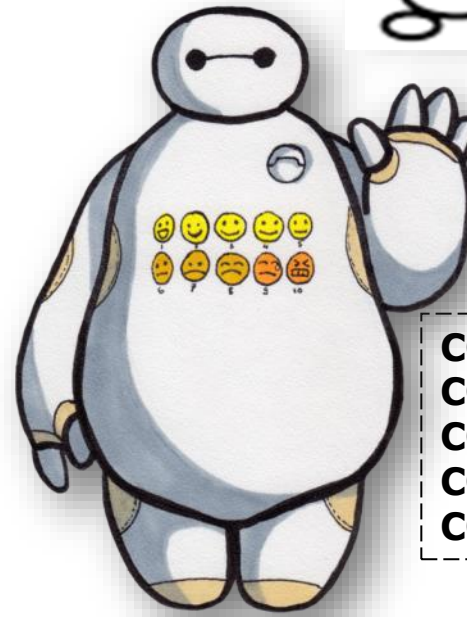
Part IV: Search techniques

- Depth First, Breadth First, UCS, A*
- Game Playing (MiniMax, Alpha-beta Pruning)



AI @ the School – Courses

- ▶ **Part I: Fundamental Issues**
 - Combinatorial explosion, experiments
- ▶ **Part II: Machine learning**
 - Artificial Neural Networks
 - Regression
 - Decision Tree
- ▶ **Part III: Intelligent Systems**
 - KRR, Bayes Rule
 - NLP, LLMs
- ▶ **Part IV: Search techniques**
 - Depth First, Breadth First, UCS, A*
 - Game Playing (MiniMax, Alpha-beta)



COMP2001 - AI Methods
COMP2005 - Image Processing
COMP3007 - Computer Vision
COMP3004 - Design Intelligent Agents
COMP3008 - Symbolic AI
COMP3009 - Machine Learning
COMP3074 - Human AI Interaction

COMP1008 - Fundamentals of AI
COMP3003 - Individual Project

COMP4030 - Data Science and Machine Learning
COMP4033 - Fuzzy Sets and Fuzzy Systems
COMP4041 - Linear and Discrete Optimisation
COMP4124 - Big Data Learning and Tech
COMP4034 - Robotic Systems

AI @ the School – Research

School of Computer Science

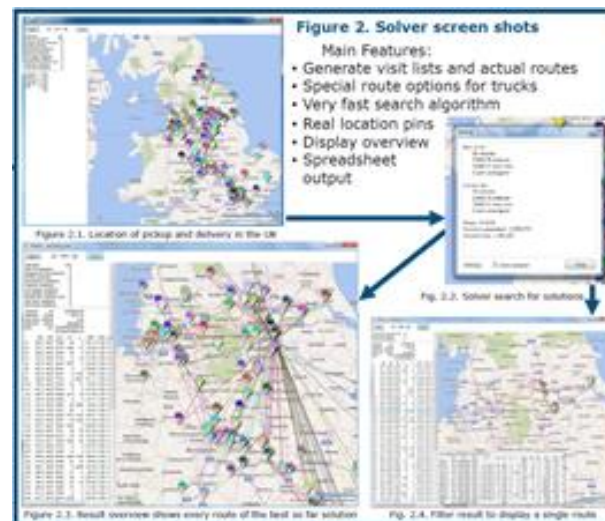
Research groups



Mixed Reality
Laboratory (MRL)



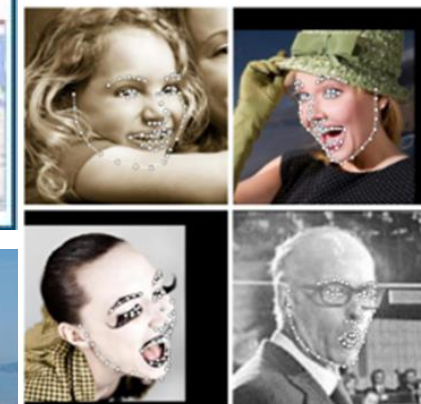
Cyber Security
(CybSec)



Computational
Optimisation and
Learning (COL)



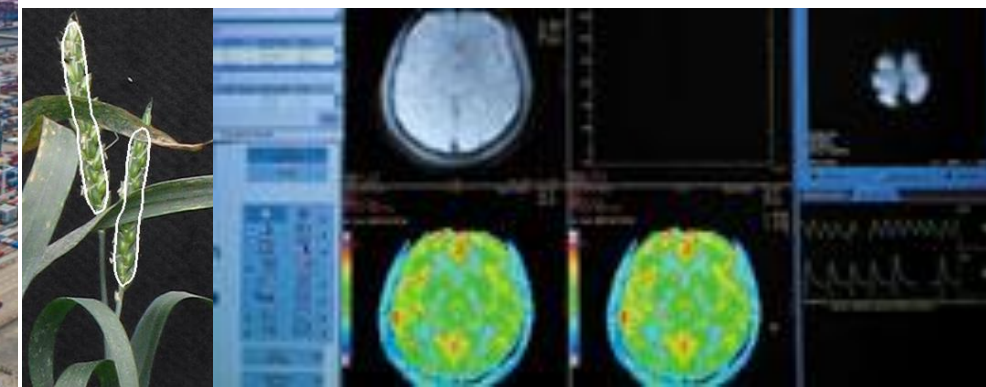
Functional
Programming Lab
(FPL)



Computer Vision
Laboratory (CVL)



CHART



AI – Extensions

- ▶ If you'd like to learn more about machine learning and AI

coursera Explore ▾ What do you want to learn? 🔍

Home > Browse > Data Science > Machine Learning

Stanford | DeepLearning.AI

Machine Learning Specialization

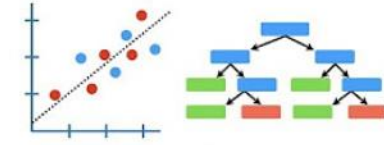
#BreakIntoAI with Machine Learning Specialization. Master fundamental AI concepts and develop practical machine learning skills in the beginner-friendly, 3-course program by AI visionary Andrew Ng

Instructors: [Andrew Ng](#) +3 more [Top Instructor](#)

Start free trial
Starts Mar 19

591,646 already enrolled

A Gentle Introduction to....



...Machine Learning!!!

Machine Learning

by StatQuest with Josh Starmer

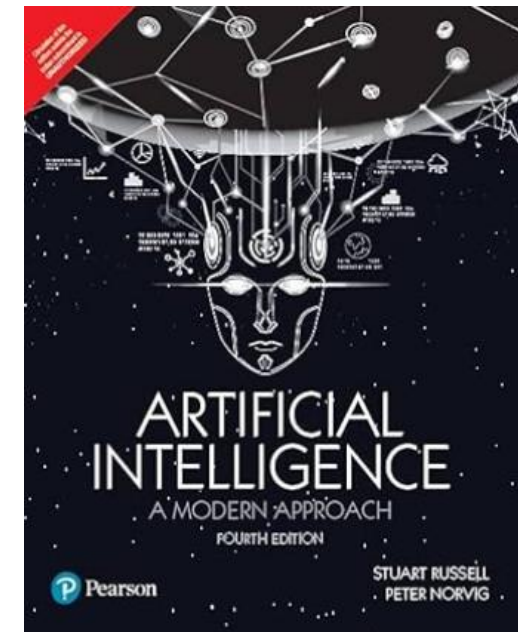
Playlist • 100 videos • 5,209,135 views

Machine Learning covers a lot of topics and this can be intimidating. However, there is no reason to fear, the ...more

▶ Play all



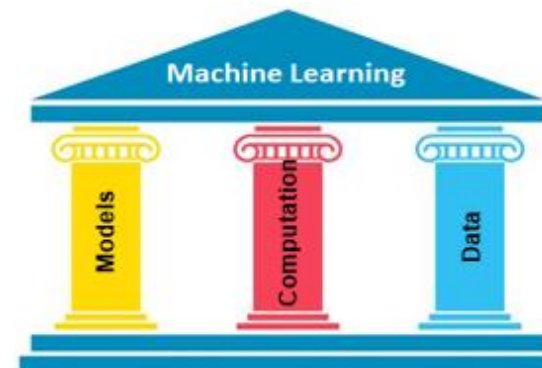
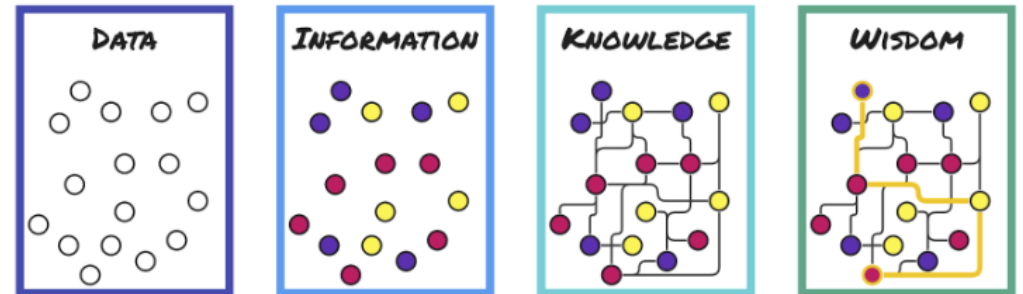
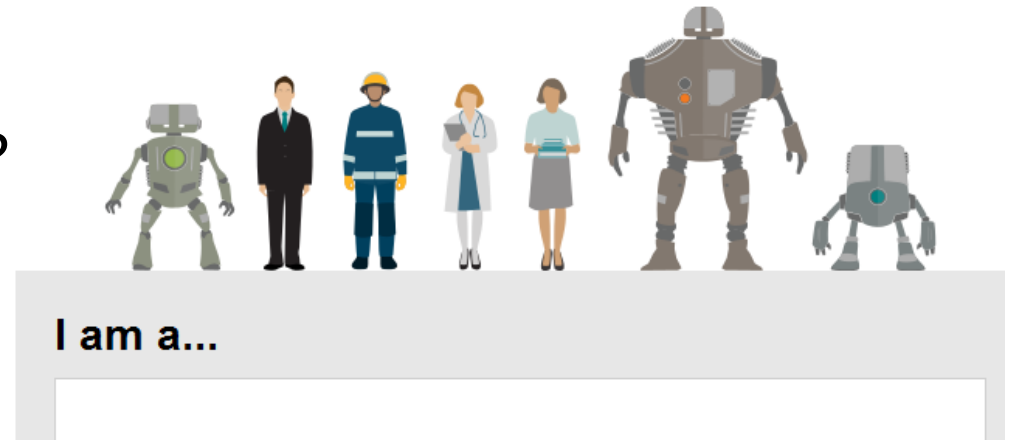
Jake VanderPlas



AI – What's Next?

About 35% of current jobs in the UK are at high risk of computerisation over the following 20 years, according to a study by researchers at Oxford University and Deloitte.

- ▶ Replace human, jobs or (which) parts of jobs?
- ▶ Or automation rather than replacing?
- ▶ How is AI used in everyday life?
 - Science, Marketing, Game, Education, ???
- ▶ Issues!? e.g. ethical
 - Data: degradation, contamination, loss of diversity and novelty
 - Algorithms / model (efficiency)
 - Computation
- ▶ Your perspective?



COMP1008 – Assessment

- ▶ Exam (75% in COMP1008)
 - [ExamSys](#), time TBC
 - No coding required
 - Five parts (75 marks in total): AI basics, Intelligent Systems, Machine Learning, Tree Search, Game Playing
 - All lecture content, unless otherwise specified
 - [COMP1008 Sample Exam](#) (suggested answers after submission)
 - Exam questions in lecture slides